

CareMate Component Interaction Diagram

Architecture and feature-flow diagram for Ayaansh's YICTE project. It shows how the Raspberry Pi robot, Pico microcontroller, sensors, actuators, cloud AI services, Windows laptop, TV, speaker/microphone and mobile face screen interact.

Scope: This diagram is based on the project component list and functionality description provided for CareMate. It labels interaction types such as Wi-Fi, Bluetooth, USB/COM serial, GPIO/PWM/sensor wiring, HDMI, API calls and power.

1. Interaction Legend

☐ Wi-Fi / socket commands

☐ Bluetooth audio

☐ USB / COM serial

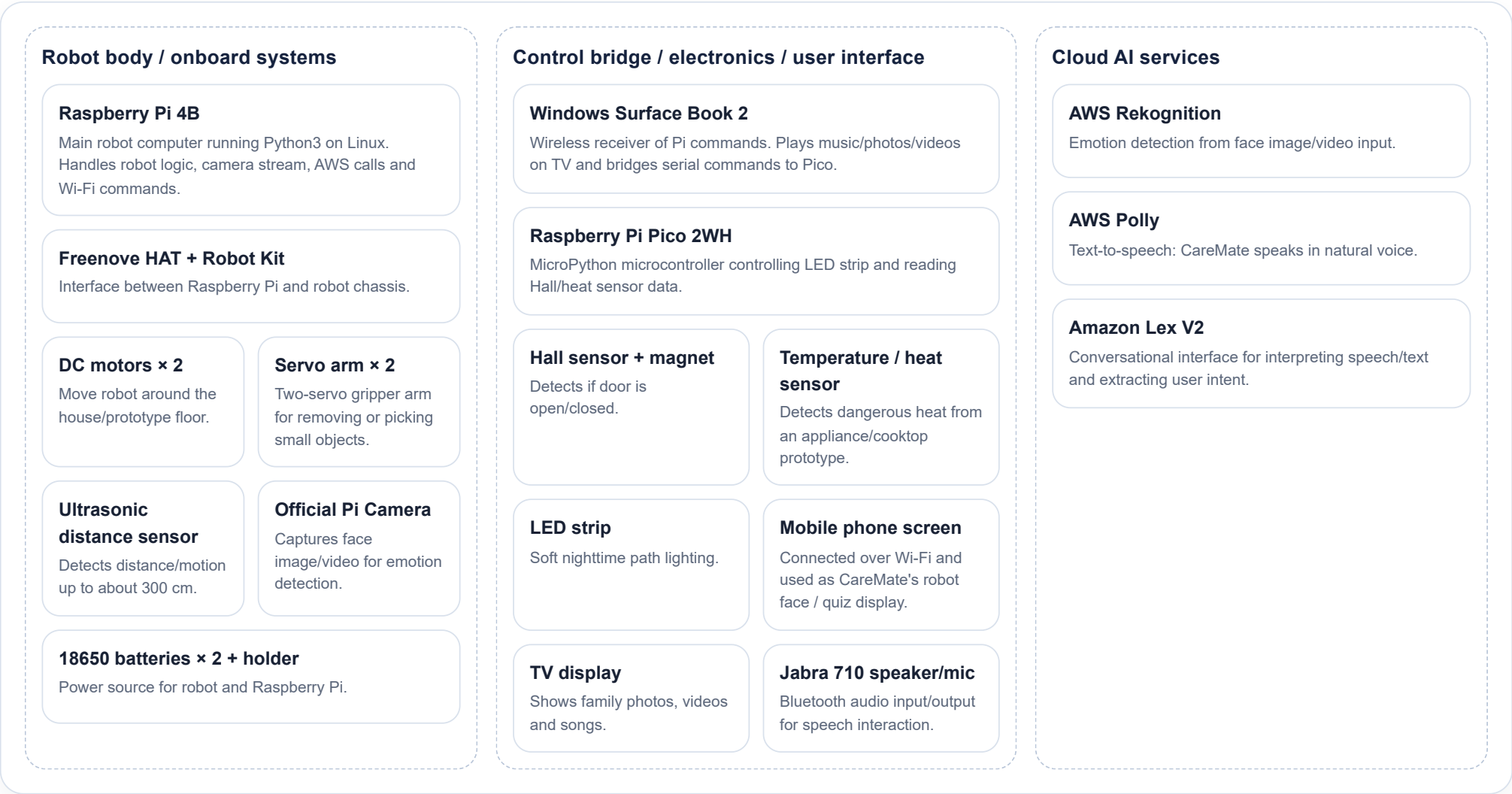
☐ GPIO / PWM / sensor wiring

☐ HDMI display

☐ Internet API call

☐ Battery power

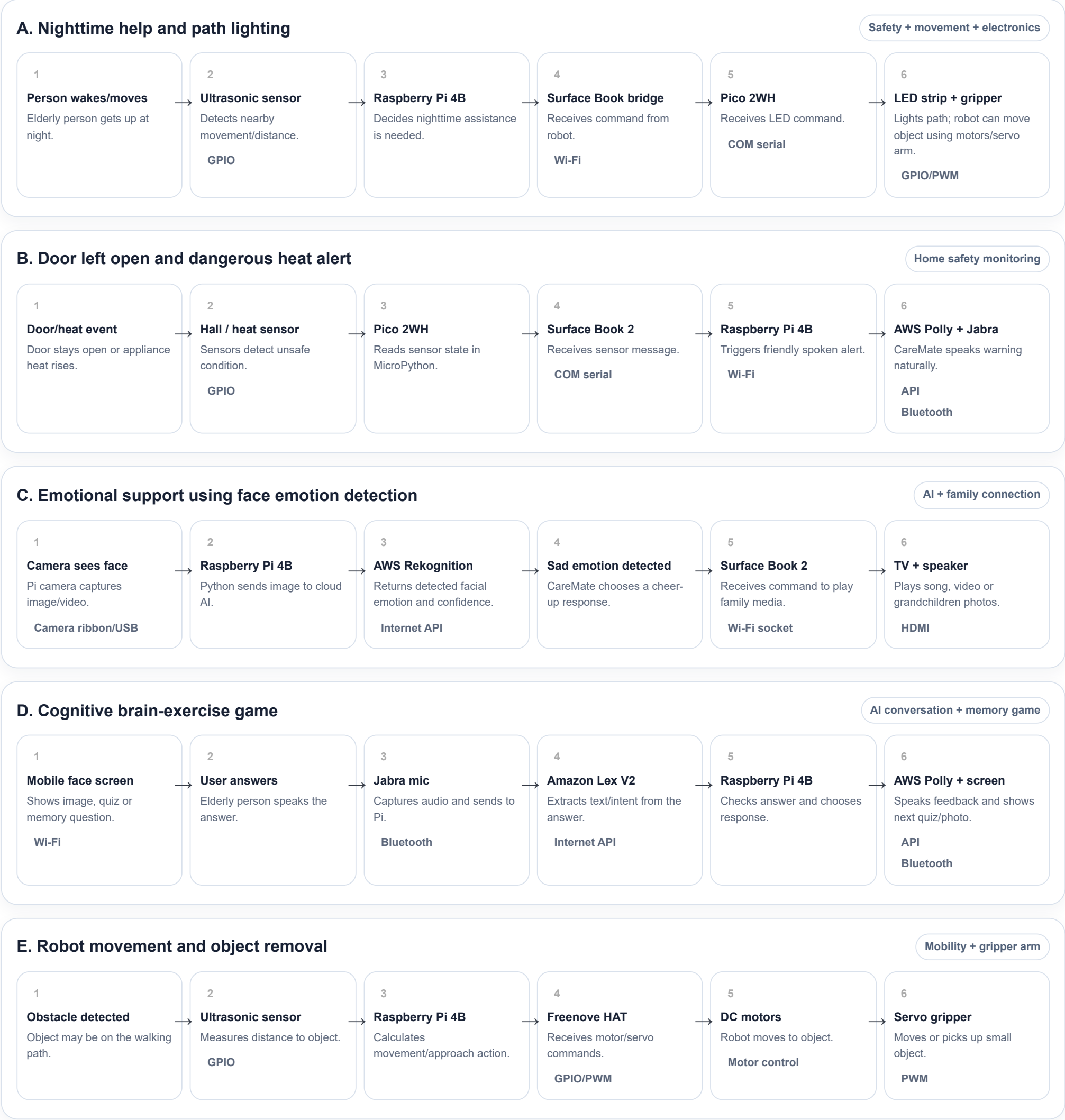
2. High-Level Component Map



3. Connection Summary

From	To	Interaction type	Purpose
Raspberry Pi 4B	Surface Book 2	Wi-Fi socket	Robot sends commands to play music, videos or photos on the TV.
Surface Book 2	Raspberry Pi Pico 2WH	USB / COM serial	Laptop sends LED control commands and receives sensor status.
Pico 2WH	LED strip	GPIO / PWM	Turns on soft path lighting for nighttime assistance.
Hall sensor + heat sensor	Pico 2WH	GPIO / sensor wiring	Detects open door and dangerous heat levels.
Surface Book 2	TV	HDMI	Displays family photos/videos and plays content for emotional support.
Jabra 710	Raspberry Pi 4B	Bluetooth	Microphone and speaker for CareMate conversation.
Raspberry Pi 4B	AWS Rekognition	Internet API	Sends camera images/video for face emotion detection.
Raspberry Pi 4B	AWS Polly	Internet API	Converts CareMate's text alert into speech audio.
Raspberry Pi 4B	Amazon Lex V2	Internet API	Understands spoken quiz responses and extracts intent.
Raspberry Pi 4B	Freenove HAT	GPIO / robot HAT	Controls robot chassis, motors and servo arm through robot interface.
18650 batteries	Robot + Raspberry Pi	Power	Powers the mobile robot and onboard computer.

4. Functionality Flow Diagrams



5. Simple Architecture Statement for Report

CareMate uses a two-computer and microcontroller architecture. The Raspberry Pi 4B is the robot brain for Python logic, camera input, AWS AI services and movement decisions. The Windows Surface Book 2 acts as a media and electronics bridge, playing family videos/photos on the TV and exchanging serial commands with the Raspberry Pi Pico 2WH. The Pico handles low-level electronics such as LED strip control and safety sensors. AWS Rekognition, Polly and Lex V2 provide emotion detection, natural speech and conversational intent understanding.